$T(n) \in \Theta(n) \iff \exists c>0, n_0$ s.t. $\forall n \geq n_0 \quad T(n) \leq c \cdot n$

Workshop question

$T(n) = 2n + 3 \in \Theta(n)$

zz: $2n+3 \leq c \cdot n$ 1 Gleichung, 2 unbekannte

1) Rearrange to isolate n
2) Freely choose c & read off (calculate n_0)

there might be some restriction for c that you can read off the equation

1) $2n+3 \leq c \cdot n \quad | -2n$

$3 \leq (c-2) \cdot n \quad | : (c-2)$ (fine because $c>0$)

$\frac{3}{c-2} \leq n$

$\hookrightarrow c > 2$

choose $c=3$ $\quad \frac{3}{3-2} = 3 \leq n \rightsquigarrow c=3, n=3$

c fixed, for any larger n the inequality only gets looser

Practice exam question

$T(n) = n(n+1)$

show $T(n) \in O(n^2)$

To show: $\exists c>0, n_0$ s.t. $T(n) \leq c \cdot n^2$

$n(n+1) \leq c \cdot n^2$

$n^2 + n \leq c \cdot n^2 \quad | -n^2$

$n \leq (c-1) \cdot n^2 \quad | : n$

$1 \leq (c-1) \cdot n \quad | : (c-1)$

$\frac{1}{(c-1)} \leq n$

$\rightsquigarrow c > 1$ choose $c=2$

$\frac{1}{1} \leq n \rightsquigarrow n=1$

Verify: $1(1+1) \leq 2 \cdot 1^2$

$2 \leq 2$ ✓

```
int iterations = 0;
for (int i=1; i≤n; i*=2) {
    for (int j = 1; j≤i ; j++) {
        iterations++;
}}
```

Workshop question

inner loop iterates i times

$\rightsquigarrow 1+2+4+\ldots+n$ iterations

so $2^0 \; 2^1 \; 2^2 \ldots 2^k$ iterations

$\downarrow$

$=n$, so runs until $k = \log_2(n)$

use: $\sum_{i=0}^{k-1} a \cdot r^i = \frac{a(1-r^k)}{1-r}$

$$\text{iterations}(n) = \sum_{k=0}^{\log_2(n)} 2^k = \frac{1-2^{\log_2(n)+1}}{1-2}$$

$$= \frac{1-2 \cdot 2^{\log_2(n)}}{-1} = \frac{1-2n}{-1} = 2n-1 \in \Theta(n)$$